IOT-BASED LABORATORY EQUIPMENT TRACKING FOR COMPUTING LABORATORIES

Concept Paper — Revised (version 2)
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Revisions in this version: the problem statement was narrowed to the setting of the study, and the objectives were rewritten to be measurable, as asked by the Instructor 1 and the Adviser on the first version.

Background of the Study

The computing laboratories lend projectors, development boards, networking kits and spare peripherals to classes every day. Borrowing is recorded on paper, items are moved between rooms without a record, and the semestral inventory regularly finds equipment missing or in the wrong room.

Bluetooth Low Energy (BLE) tags are inexpensive and last months on a coin cell. Gateways in each room can detect the tags and report their signal strength, from which a server can estimate the room an item is in. Combined with a borrowing log, such a system would show where every item is and flag items that leave the laboratories.

Statement of the Problem

Laboratory staff cannot locate equipment without searching every room, borrowing records are incomplete, and losses are discovered only at inventory time.

Objectives

The study aims to design, build and evaluate an IoT system that locates tagged laboratory equipment by room and records its movement. Specifically, it aims to:

1. determine the equipment-handling problems of the computing laboratories through observation and interviews;

2. design a BLE tag and gateway network and a room-level location algorithm based on received signal strength;
3. develop a web dashboard that shows the location and movement history of each item and alerts staff when an item leaves the laboratories;
4. measure the room-level location accuracy of the algorithm in the four computing laboratories; and
5. evaluate the system using ISO/IEC 25010 with laboratory staff and IT experts.

Scope and Delimitations

The study covers the four computing laboratories and the equipment room on one floor, using BLE tags on 60 items and one gateway per room. Location is resolved to the room, not to a position within it.

The study does not track people, does not cover equipment outside the floor, and does not replace the university's property inventory system; it provides location and movement records that staff can reconcile with it.

Proposed Method

The group will design the sensing network and location algorithm, build a prototype, measure its accuracy and evaluate it with ISO/IEC 25010.

Expected Output

IoT-based equipment tracker and Accuracy and evaluation results.

References

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Dummy concept paper for testing the HAU-SOC prototype. All content is placeholder.